

Horizon Score Strategy

Strategy construction, backtesting, robustness and final evaluation

Can Horizon Score turn predictions into a robust tradable strategy?

Consumes confirmed research outputs, constructs long-only portfolios from horizon-aware signals, and evaluates the resulting strategy and GBP account. Model fitting remains upstream.

10 September 2026 | Split-project edition

Adapted from Equity Selector Project Report, 4 September 2026. Updated to document the completed local project separation and its shared model/data boundary.

This is methodology and implementation documentation. It does not present a new empirical performance study.

Contents

1 Executive summary and project boundary	3
2 Research inputs, chronology and frozen identity	3
3 Horizon Score research	4
4 Prediction cache and strategy-facing data	5
5 Signal construction, allocation and accounting	6
6 Portfolio simulations and candidate selection	7
7 Precise reconstruction and robustness tests	8
8 Comprehensive post-backtest evaluation	9
9 Macro-regime, rolling and calendar diagnostics	11
10 GBP account checks and frozen final evaluation	12
11 Storage, execution sequence and validation evidence	13
12 Signal-group reference and document provenance	15

Reading this report

Numbered sections describe this project. The interface section identifies the artifacts shared with the companion project. Original methodology is retained where relevant; changes introduced by the split are explained explicitly.

Source and scope

Historical example settings from the original report are distinguished from the editable settings inspected in the refactored code. The root launchers remain authoritative for any actual run.

1 Executive summary and project boundary

Horizon Score Strategy is the downstream project that converts confirmed target predictions into a repeatable portfolio-allocation method. It researches horizon weights, creates prediction caches, compares portfolio settings, reconstructs selected candidates and carries frozen choices into a later evaluation period.

The project preserves the original signal transformation, positive-score allocation, timing and trading-cost conventions. It adds a comprehensive evaluator operating from daily returns and ticker weights, together with shared macro regime definitions produced by Equity Model Research. Rerun-based robustness checks remain separate and intact.

Owned by Horizon Score Strategy	Supplied by Equity Model Research
Horizon weights and signal-group portfolio construction	Confirmed target/model metadata and selected feature definitions
Prediction consumption and strategy caches	Fitted estimators, scalers/encoders and provenance manifests
Portfolio simulation, robustness and precise backtests	Prepared stock/macro data and target-related information
GBP account checks and frozen final validation	Macro_Regimes.txt with shared frozen macro definitions

This report describes a research method and its implemented interfaces. No live deployment, new return study or investment-performance claim is implied by the project separation.

```
Projects/Horizon Score Strategy/
Run Strategy.py Prepare Strategy.py
Horizon Score Backtests.py Backtest Database.py
Backtest Simulations.py Precise Backtest.py
Evaluate Backtest.py
GBP Portfolio Check.py Frozen Final Test.py
horizon_score/ tests/ docs/
```

2 Research inputs, chronology and frozen identity

The upstream output contract

Selected models, target metrics and serialised parameters arrive through Final_Test_Results.db. Selected_Features.txt supplies the universe/target feature map; Features_Targets_Data.db supplies prepared research inputs where needed. Fitted artifacts are loaded through equity_selector.artifacts.load_models_and_predictions. Macro_Regimes.txt supplies inclusive date intervals for Growth, Inflation, Labour, Liquidity, Consumer and Housing.

Target metadata includes statistical type, economic role, horizon, Predictability Score and Quality Score. Those quantities describe the upstream model; Horizon Score and Type Score determine its downstream influence. The strategy does not repeat target screening or model search.

Strategy phase	Historical role / inspected dates
Horizon research	2016-2018; earlier research observations define the required fits
Portfolio selection cache	Training through 31 December 2018; predictions 1 January 2019-30 December 2022

Strategy phase	Historical role / inspected dates
Final cache and frozen evaluation	Training through 31 December 2022; predictions 1 January 2023-31 August 2026

Dates above describe the existing historical research profile. Target-specific purges and availability calendars still govern the exact fit boundary. The later final-cache fit is explicitly produced upstream; the frozen final evaluator itself does not fit models or search for a new portfolio.

No silent model refit

Previously, Horizon and cache stages fitted selected models internally. A missing fit now produces batched research requests and stops before portfolio construction. Produce Confirmed Models.py runs in the upstream project, then the strategy resumes. Request identities cover the exact training data, selected metadata, feature order and purge policy; no BACKTEST rows enter fitting requests.

Frozen inputs take precedence

Phase-staged Final_Test_Results.db, Selected_Features.txt and Macro_Regimes.txt are preferred over live research copies. This prevents a later research update from silently replacing a frozen selection. Horizon configurations are ordered; Simulation ID, Horizon Score Index and complete portfolio settings must remain associated throughout the research record.

3 Horizon Score research

Predict once, vary horizon influence

Horizon Score Backtests.py loads confirmed historical fits, generates the later target predictions and reuses that common prediction panel while varying target/horizon weights. A zero horizon weight disables a component; positive values control its contribution relative to the other represented components. Only retained target/horizon pairs enter the active search.

The research question is whether short-, medium- or longer-horizon information is useful within the selected target types. Retained mappings become inputs to the later portfolio-setting sweep, not a new model selection step.

Original illustrative grid and inspected implementation

Original September 4 example	Interpretation
ALPHA: 20d and 60d	Weights 0.0, 0.5, 1.0
RELATIVE_ALPHA: 20d and 60d	Weights 0.0, 0.5, 1.0
DIRECTION: 20d; DOWNSIDE: 20d	Weights 0.0, 0.5, 1.0
Six variable components	$3^6 = 729$ initial combinations in that example

The 729-combination example is preserved as a historical design illustration, not asserted as the current launcher grid. The refactored launcher contains a broader candidate catalogue and nonempty random-screen rounds: (20, 0.15), (50, 0.15), (100, 0.20). The maximum exhaustive survivor count is 1,000. Inspect the active HORIZON_SCORE_VALUES and screening schedule before a run.

Portfolio backgrounds and economic comparison

The original report illustrates 27 Cartesian backgrounds from three signal-group configurations, three weight caps and three concentration settings. Horizon-screening backgrounds are independent of the later portfolio-simulation catalogue. Each comparison uses matched inputs and preserves the same allocation equations.

Benchmark-relative Backtest Quality

The historical Horizon criterion combines relative total return, Sharpe, maximum drawdown and average drawdown. Return and Sharpe improvements receive positive values, while drawdown components reward smaller magnitudes. The strategy and benchmark are compared over the matching evaluation dates.

```
q_return = (R_strategy - R_market) /
(abs(R_strategy) + abs(R_market) + epsilon)
q_drawdown = (abs(D_market) - abs(D_strategy)) /
(abs(D_market) + abs(D_strategy) + epsilon)
BQ = 0.35*q_return + 0.25*q_Sharpe
+ 0.25*q_max_drawdown + 0.15*q_average_drawdown
```

This is the original report's Horizon-score explanation. Existing stage-specific selection-quality conventions and compatibility adapters remain in the code; the new post-backtest evaluator does not replace the historical selection objective or silently re-rank strategies.

Sensitivity, screening and survivor search

Sensitivity tests vary one horizon component and compare the change in quality. The original reference uses a 0.003 freeze range and 0.002 near-best tolerance. Configured random-context rounds reduce the candidate lists; exhaustive search runs only once the surviving product is within its configured bound. Similarity checks use the original 0.95 configuration-correlation reference. Strong retained mappings are saved in fixed order to Top_Horizon_Scores.txt.

4 Prediction cache and strategy-facing data

Backtest Database.py rebuilds the required stock/market feature inputs for its selected historical phase, loads the already-produced fitted models and caches later predictions/signals. It does not fit an estimator. Feature warm-up and target look-ahead requirements retain the original date handling, while selected PIT macro predictors are attached from the shared existing database by backward availability date.

CACHE_PHASE selects the selection or final window. Separate phase directories prevent the final cache from overwriting selection data. The benchmark is cached alongside stock inputs, so later evaluations can use the same market reference without repeating model or data preparation.

Cached table / field	Role
Stocks	Long-form prepared stock signal panel
Market	Corresponding benchmark inputs
Date and Ticker	Observation/security identity
Return	Realised return for subsequent portfolio calculations
Portfolio Target Type	Economic signal category
Horizon Key	Connection to the frozen horizon mapping

Cached table / field	Role
Signal	Prepared target-type/horizon signal
Passed Strategies	Selection-phase rows with exact Simulation IDs and settings

The prepared cache is not the evaluator's daily-weight input. The portfolio engine first converts cached signals into one realised return and one weight per ticker per date. The post-backtest evaluator then analyses that resulting stream.

5 Signal construction, allocation and accounting

Contributions and refresh

The portfolio stage joins each cached signal to its Horizon Score and Type Score. The Type Score is supplied by the broad group configuration: Ranking, Direction, Risk, Opportunity or Special. Each signal contributes according to both its horizon and its group weighting.

$Contribution_{i,j,t} = Signal_{i,j,t} \times HorizonScore_j \times TypeScore_j$ The refresh rule holds each target/horizon contribution for its configured number of observations before accepting the next signal value. Contributions are aggregated by stock and date. Direction information supplies an additional long-only gate: a negative aggregate direction makes the resulting stock score non-positive.

```
Contribution[i,j,t] = Signal[i,j,t] * HorizonScore[j] * TypeScore[j]
Refresh observations = max(1, ceiling(multiplier * horizon))
```

Refresh holds each target/horizon contribution within its stock/type/horizon series until the next refresh observation. The multiplier acts on observation counts, not arbitrary calendar gaps. Contributions aggregate by stock and date, with the direction gate applied before long-only allocation.

Positive-score portfolio weights

Only positive stock scores enter the long-only allocation. Scores are scaled and normalised, then blended with an equal-weight component using the concentration setting. A capped redistribution assigns the available capital while respecting the maximum weight per stock.

$u_i = (1 - \lambda) s_i / \sum s_j + \lambda / n$ Here s denotes a positive stock score, n is the number of eligible stocks and λ is the concentration setting. A value of zero gives score-proportional starting weights; a value of one gives equal starting weights. The cap is applied through successive redistribution of the remaining allocation.

$0 \leq w_i \leq w_{max}, \sum w_i \leq 1$ When the eligible positions do not use the full capital allocation, the remainder is held as cash. Each portfolio result records the stock weights used to earn the next observation's returns.

```
u[i] = (1 - concentration) * score[i] / sum(score)
+ concentration / number_of_positive_scores
0 <= weight[i] <= maximum_weight; sum(weights) <= 1
```

The concentration setting blends score-proportional and equal starting weights. Capped redistribution assigns the remaining investable capital. Unused capital is cash; the code does not fabricate an extra positive-score position to reach full exposure.

Timing and transaction costs

Signals available at observation t determine weights that earn returns at the next observation. Each saved daily row records the weights actually used for its realised return. Costs depend on changes in target weights, including initial entry from cash.

```
traded_notional[t] = sum(abs(weight[t] - weight[t-1]))
reported_turnover[t] = 0.5 * traded_notional[t]
gross_return[t+1] = sum(weight[t] * asset_return[t+1])
net_return[t+1] = gross_return[t+1] - cost * traded_notional[t]
```

The original formula labelled full traded notional as Turnover. The new evaluator makes the distinction explicit: reported turnover is half the absolute weight change, while fees remain charged on the full buy-plus-sell amount. This clarification preserves the original return calculation.

6 Portfolio simulations and candidate selection

Inspected portfolio-setting sweep

Dimension	Refactored launcher values
Refresh multiplier	0.0, 0.25, 0.5, 0.75, 1.0
Concentration setting	0.0, 0.05, 0.1, 0.2, 0.3
Maximum stock weight	0.10, 0.15, 0.20, 0.25, 0.30
Signal groups	Unchanged simulation catalogue in horizon_score/groups.py
Horizon mappings	Each retained mapping from Top_Horizon_Scores.txt

The September 4 source report showed a narrower illustrative sweep (caps 0.05/0.10; concentration 0/0.25/0.50; refresh 0.5/1.0). Those values are not substituted for the inspected launcher. The root settings and saved rows define an actual run.

Selection evidence and similarity

The selected stock universe supplies the principal portfolio test. Additional stock-removal and comparison-universe analyses investigate whether the strategy's result is broadly supported across its security inputs. The same signal and allocation rules are used when making those comparisons.

Each simulation is identified by a Simulation ID. Its stored settings include the horizon configuration index, signal-group configuration, refresh multiplier, maximum weight and concentration penalty. Result fields describe return, Sharpe, drawdown, benchmark-relative quality and the applicable robustness measurements.

Portfolio_Simulation_Results.db contains the experiment results used to compare candidates and benchmark behaviour. The selected configurations are carried forward with their full setting identity so that the precise evaluation can reconstruct the same allocation rule.

Candidates are compared using benchmark-relative performance, target-role evidence, parameter stability and stock-input transfer. The original similarity references are return correlation 0.95, relative performance difference 0.10 and average stock overlap 0.80. Selected rows carry their full settings rather than only a descriptive name.

- Simulation ID and ordered Horizon Score Index
- Type Configuration and its exact group values
- Rebalance Multiplier, Max Weight and Concentration Penalty
- Original strategy/benchmark metrics and relevant robustness evidence
- Daily cached inputs and the recorded selection entering detailed evaluation

Configuration consistency after the split

A pre-existing mismatch meant the simulation launcher could select group names that precise/frozen launchers could not reconstruct. The split centralizes the unchanged simulation groups and makes validators accept those groups plus legacy frozen names. It does not alter the simulation search space. Horizon-screening backgrounds remain independent.

7 Precise reconstruction and robustness tests

Precise Backtest.py reconstructs the selected portfolio configurations from the cached signals and their saved settings. It checks the resulting metrics against the saved simulation quantities and produces a detailed record for each candidate.

The exact configuration includes the Horizon Score Index, Type Configuration, Rebalance Multiplier, Max Weight and Concentration Penalty. The script uses the same shared signal transformation and portfolio allocation functions as the simulation layer.

The detailed output associates each diagnostic with its Simulation ID and saved row. Daily streams, neighbours, stock removals and other intermediate results can therefore be traced back to the same candidate. These records support subsequent analysis without requiring the researcher to infer which settings produced a chart or summary.

Retained diagnostic	What it establishes
Saved-metric reconstruction	The selected settings reproduce their stored research identity
Best day/week/month/year removal	Sensitivity to unusually favourable return periods
Rolling 252-observation quality	Behaviour of successive annual sections
Leave-one-stock-out tests	Dependence on individual security inputs
Nearby-parameter tests	Local behaviour around refresh, concentration and position-cap settings
Unseen-stock comparison	Transfer to a disjoint comparison universe
Cost scenarios	Sensitivity to actual traded notional and execution allowances
PSR and DSR	Distributional confidence and multiple-testing calibration, subject to their assumptions

Neighbourhood, stock-removal, unseen-stock and frozen validation checks inherently need the strategy to be rerun. They are retained. General statistics obtainable from daily returns/weights are supplied by the canonical evaluator, avoiding independent Sharpe, expected-shortfall and cost implementations in the precise layer.

Reference assumptions and stored evidence

Precise research setting	Reference
Annualisation / annual reference rate	252 observations / 0.0
Random seed	42
DSR calibration sample	Up to 1,000 sampled configurations from the saved grid
DSR trials	Saved unique grid count, or explicit research count
Execution-cost fraction in the precise launcher	0.0005 of full traded notional
Rolling quality diagnostic	252 observations

Final_Strategy_Evaluation.db and companion CSV/pickle outputs retain the reconstructed summary, daily streams, correlation matrix, DSR calibration and per-strategy diagnostics. Evaluation_Assumptions.json records the legacy precise-stage assumptions. Each finalist now also receives its comprehensive post-backtest report.

8 Comprehensive post-backtest evaluation

Inputs, date alignment and standalone use

evaluate_backtest(backtest_results, regimes, market, ...) operates without predictive-model calls.

backtest_results contains Date, gross Return and one daily portfolio-weight column per ticker. Weights must be finite and in [0, 1]; invested gross exposure is normally approximately one, with cash allowed. Duplicate dates, invalid returns and overlapping regime intervals are rejected.

Benchmark input supports Return, Adj Close or Close and ordinary single-ticker yfinance MultiIndex output. Dates are normalized, sorted and joined explicitly. Price returns use no filling; missing benchmark dates are not matched by row position. Statistics requiring the market use the intersection of valid strategy and benchmark observations.

```
python "Projects/Horizon Score Strategy/Evaluate Backtest.py" \
--backtest /path/to/daily_weights.csv \
--market /path/to/sp500.csv \
--output-dir /path/to/final_evaluation
```

Overall strategy metrics

Metric group	Measures
Return and risk	Observations; Total Return; CAGR; Annualised Volatility; Sharpe Ratio; Sortino Ratio; Calmar Ratio
Drawdown and recovery	Max, Average and Current Drawdown; Longest Drawdown Duration (observations); Longest Drawdown Days; Longest Recovery Period (calendar days for completed recoveries)
Tail risk	VaR 95%; Expected Shortfall 95%, expressed as signed returns
Daily outcomes	Positive/Negative Day Rate; Average Positive/Negative Day; Profit Factor; Best/Worst Day
Distribution and confidence	Daily Skew; Daily Excess Kurtosis; Probabilistic Sharpe Ratio

Portfolio and weight metrics

Metric group	Measures
Holdings and exposure	Average, median, minimum and maximum holdings; average/minimum/maximum gross weight
Position concentration	Average largest position; maximum position observed; average/median HHI; average/median/minimum effective holdings
Turnover and changes	Average/median/p95/maximum daily turnover; annualised turnover; fraction of days with portfolio changes
Entries, exits and reweighting	Average entries/exits per day; maximum entries/exits in one day; average absolute weight change

```
HHI[t] = sum(weight[i,t]^2)
EffectiveHoldings[t] = 1 / HHI[t], if HHI[t] > 0
AnnualisedTurnover = mean(daily_turnover) * 252
```

HHI uses raw weights, not a renormalized invested sleeve. Fully cash days have undefined effective holdings. Initial entry counts in traded notional and entries. Average absolute weight change averages absolute ticker-weight changes, whereas turnover sums them before applying its one-half convention.

Transaction-cost metrics

The cost panel reports Trading Cost Fraction, Trading Cost bps, Gross Total Return, Net Total Return, Net CAGR, Gross Sharpe, Sharpe After Costs, Sharpe at 2x Costs, Net Max Drawdown and Break-Even Cost bps. Net returns subtract the configured fee from full traded notional. Break-even cost is the fee at which compounded net wealth reaches its starting value; its zero/no-trading cases are explicit.

The reusable evaluator defaults to a 0.001 fee fraction; the precise strategy launcher passes its existing 0.0005 research allowance. Cost-stress insolvency is reported as an error rather than as an ordinary return statistic. Input Return must be gross of the fees being evaluated.

Benchmark metrics

Group	Measures on aligned observations
Coverage and comparative performance	Aligned Observations; strategy/market CAGR, annualised volatility, Sharpe and maximum drawdown
Dependence and active risk	Strategy-market correlation; beta; annualised alpha; tracking error; information ratio; annualised active return
Market-direction response	Upside Capture; Downside Capture; mean strategy return on positive-market and negative-market days

Alpha uses arithmetic excess daily means and beta, annualised by 252. Tracking error is the annualised standard deviation of active daily returns. Capture ratios use conditional arithmetic means rather than an unstated geometric convention. The annual risk-free rate defaults to zero and is converted to a daily rate when used.

9 Macro-regime, rolling and calendar diagnostics

Shared frozen regimes

Regimes are produced upstream from existing Macro PIT features; the strategy loads the frozen dictionary with `ast.literal_eval`. It does not recalibrate macro thresholds to improve a strategy result. The six families and their active states are retained as separate conditioning dimensions.

Family	States, with Unknown also retained
Growth	Expansion, Neutral, Contraction
Inflation	Rising, Stable, Falling
Labour	Strong, Neutral, Deteriorating
Liquidity	Expanding, Neutral, Contracting
Consumer	Improving, Neutral, Weakening
Housing	Improving, Neutral, Weakening

Every family/state receives an aggregate row, including empty and Unknown states. Rows report days and backtest fraction; return, volatility, Sharpe, Sortino, tail risk and positive-day rate; holdings, position size, HHI, effective holdings and turnover; and available aligned benchmark, beta/alpha and active-risk statistics. Dates outside supplied intervals are labelled Unknown.

Disconnected episodes are not a continuous investment

A regime can recur in separated historical episodes. Aggregate return and Sharpe describe exposure to that state, but a drawdown on the stitched sequence would describe a fictitious investment path. Aggregate regime CAGR therefore uses exposure observations and is labelled accordingly. Aggregate path-dependent drawdown and Calmar fields are undefined; contiguous episode rows retain meaningful drawdown/recovery diagnostics.

Rolling and calendar outputs

Output	Measures
63, 126 and 252 observations	Rolling Return; Annualised Return; Volatility; Sharpe; Market Correlation; Beta; Average Holdings; Average Turnover; Effective Holdings
Yearly performance	Strategy and market Return, Sharpe and Max Drawdown, with other available annual statistics
Monthly return table	Compounded strategy return by year and month
Drawdown episodes	Peak, start, trough, end and recovery dates; depth; durations; unrecovered status

A rolling statistic is populated only when its required window is available. Benchmark-based rolling metrics require aligned observations and do not fill missing market dates. An incomplete drawdown does not receive an invented recovery date.

Return structure and saved artifacts

```
{ 'summary', 'benchmark', 'portfolio', 'costs',
  'regimes', 'regime_episodes', 'rolling', 'yearly',
  'monthly', 'drawdowns', 'daily' }
```

Each output is a Series or DataFrame and is saved separately as CSV. The daily table retains original weights and returns and adds net returns, holdings/exposure, traded notional, wealth, drawdown, benchmark returns and regime labels. Conventions.txt and Assumptions.json record interpretation and run settings. A literal Macro_Regimes.txt snapshot records the exact supplied regime definitions for the evaluation.

10 GBP account checks and frozen final evaluation

Account valuation

The original account profile is a GBP-based Trading 212 Invest portfolio with GBP 4,000 initial capital, individual USD trades and a retained USD balance. GBP_per_USD.csv supplies the GBP value of one USD. The entire USD sleeve, including idle cash, is valued using date-aligned exchange rates.

```
1 + GBP_return[t] = (1 + USD_return[t]) * FX[t] / FX[t-1]
FX is GBP per USD
GBP_equity[t] = initial_capital * product(1 + GBP_return)
```

The configured initial and terminal currency conversions each charge 0.15%. Individual USD trades use zero broker commission; the separate execution allowance is 0.05% of full traded notional. GBP account calculations preserve capital-flow and FX handling separately from the USD strategy evaluator.

Drawdown acceptance and final freeze

A selection candidate passes the configured GBP account criterion when maximum drawdown is no worse than 20% from the running equity peak, including initial capital. GBP Check/GBP_Selection.db retains evaluated daily paths and a Passed Strategies table for candidates meeting the criterion.

Frozen Final Test.py reconstructs every frozen finalist on the separate final cache, with the saved horizon ordering, signal-group values, refresh multiplier, weight cap and concentration setting. Selection dates must precede final evaluation dates. The final stage does not automatically choose a new winner from the final outcomes.

Frozen input	Purpose
Selection database	Every finalist and its exact saved settings
Separate final Backtest_Database.db	Final-period stock and market signals/returns
Frozen Top_Horizon_Scores.txt	Configuration mapping and order
Final-phase GBP_per_USD.csv	Account valuation on the required dates
Frozen Macro_Regimes.txt	Common conditional-performance definitions

Final outputs and cost handling

Frozen_Final_Evaluation.db contains Daily Returns, Summary and Frozen Settings, with companion CSV and manifest. The account-selection phase records the drawdown acceptance outcome and passed candidates; the frozen final phase reports all retained finalists. Input hashes, declared/actual dates and settings support later metric replay and database checks.

The GBP and frozen launchers also produce comprehensive USD post-backtest reports. They recover gross USD returns from the existing net stream using the same full-traded-notional fee, then apply the evaluator once. This avoids double charging. GBP account outputs and drawdown gates remain intact. The reusable evaluate_frozen API can accept explicit regimes and market inputs; legacy callers without them retain the earlier account-only interface.

11 Storage, execution sequence and validation evidence

Strategy artifact	Role
Top_Horizon_Scores.txt	Ordered retained horizon configurations
Backtest_Database.db	Stocks/Market caches and selection-phase Passed Strategies
Portfolio_Simulation_Results.db	Portfolio experiments and comparison/robustness results
Final_Strategy_Evaluation.db and details	Precise reconstruction and rerun diagnostics
Per-strategy evaluation directories	Eleven CSV outputs, conventions, assumptions and regime snapshot
GBP Check/GBP_Selection.db	GBP account selection and approved candidates
Frozen_Final_Evaluation.db and manifest	Final-period outcomes, settings and provenance

Run sequence and configuration

Run Strategy.py exposes prepare, horizons, cache, simulations, precise, gbp_check and final stages. Prepare Strategy.py stages immutable research metadata/regimes and phase-specific FX history; final preparation additionally stages the frozen Horizon file. Complete upstream production requests whenever a historical fit is missing, then resume the requesting stage.

```
python -m pip install -e "Projects/Equity Model Research[test]"
python -m pip install -e "Projects/Horizon Score Strategy"

Upstream outputs: EQUITY_SELECTOR_DATA_DIR
Strategy outputs: HORIZON_SCORE_DATA_DIR
Strategy phase folders: portfolio_selection/ and final_evaluation/
```

Horizon output goes to the selection directory used by simulations. Existing private data is not moved automatically. Set environment variables or edit launcher settings to select the intended directories; never overwrite selection evidence with the final cache. Prepare/freeze final inputs and switch `CACHE_PHASE` before running final evaluation.

Refactor validation, not performance validation

The refactor validation recorded 125 passing tests and successful installation of both packages. A total of 103 modules imported with network/database access disabled; static checks found no strategy fitting calls. Across 12 allocation/fee configurations and 299 realised dates with eight tickers, daily returns and weights matched the original implementation exactly. A deterministic fitted-model artifact fixture also produced exactly the same predictions as the original fit-and-predict path.

Those checks confirm tested structural and numerical behavior. The repository did not contain production databases, fitted historical artifacts or a saved production return stream, so the complete empirical backtest could not be rerun or compared. No new strategy CAGR, Sharpe, regime-return estimates or live account results are reported in this document.

Known boundaries

PSR uses an IID assumption and needs at least 30 observations; DSR depends on its recorded trial/calibration assumptions. Disconnected regime statistics require exposure-based interpretation. Zero variance, empty states, all-cash effective holdings and incomplete recoveries produce explicit undefined values where appropriate. Legacy procedural stages and package interfaces remain for compatibility; larger redesigns are outside this split.

12 Signal-group reference and document provenance

Simulation catalogue retained by the split

Configuration	Ranking	Direction	Risk	Opportunity	Special
Balanced	0.30	0.25	0.25	0.15	0.05
Equal Weight	0.20	0.20	0.20	0.20	0.20
Core Balanced	0.35	0.30	0.25	0.10	0.00
Ranking Heavy	0.50	0.20	0.20	0.10	0.00
Ranking And Risk	0.45	0.15	0.30	0.10	0.00
Ranking And Direction	0.45	0.30	0.15	0.10	0.00
Direction Heavy	0.25	0.45	0.20	0.10	0.00
Direction And Risk	0.25	0.40	0.30	0.05	0.00
Risk Heavy	0.25	0.20	0.45	0.10	0.00
Conservative	0.30	0.15	0.45	0.05	0.05
Opportunity Heavy	0.25	0.20	0.20	0.35	0.00
Ranking And Special	0.40	0.15	0.20	0.10	0.15

Legacy names retained for frozen reconstruction

Configuration	Ranking	Direction	Risk	Opportunity	Special
Equal Weight Baseline	0.20	0.20	0.20	0.20	0.20
Ranking Focused	0.50	0.20	0.20	0.10	0.00
Risk Aware	0.35	0.20	0.35	0.10	0.00

These tables document the refactored reconstruction catalogues. The Horizon-screening background list has its own settings and is not inferred from this simulation catalogue. Saved group names and values must remain associated with the chosen Simulation ID.

Source map and companion report

Primary source: Equity Selector Project Report.pdf, dated 4 September 2026, supplied by the user. This report retains the strategy material from original sections 9-14 and the relevant architecture, storage, evidence and formula appendices. Model/data screening, fitting, confirmation, macro production and upstream metadata are documented in the companion Equity Model Research Project Report.

Implementation sources: the refactored strategy README, REFACTOR_REPORT.md, evaluation.py, groups.py, frozen.py and root launchers. New model/regime interfaces and evaluator descriptions update the combined-project documentation. Historical grid examples are labelled separately.